Digital Fundamental : DPP of Lec 11

Computer Science

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Q1 For a boolean expression consisting of n-variables, number of cells required in K Map = _____cells.

(A) n (B) n(n-1)

(C) 2^n (D) n^2

Q2 Simplify the given Boolean for using the Karnough map.

$F(x,y,z)=\sum(0,2,4,5)$

(A) xy+yz (B) xy'+x'z'

(C) xz+yz (D) x'y+yz'

Q3 Which of the following are the methods of minimizing or reducing the boolean expressions?

(A) By using the Laws of Boolean Algebra

(B) By using Karnaugh Maps

(C) Both A and B

(D) Neither A nor B

Q4 How many forms of K-map are there?

(A) 2 (B) 3

(C) 1 (D) 4

Q5 Which of the following statements is/are True for K-Map?

I. We can either group 0's with 0's or 1's with 1's but we can not group 0's and 1's together.

II. X representing don't care can be grouped with 0's as well as 1's.

III. Groups may overlap each other and a group can only contain 2^n i.e. 1, 2, 4, 8, 16, and so on number of cells.

IV. We can create groups of diagonal or any other shape.

(A) I, II, and III only

(B) I, and II only

(C) II, and IV only

(D) II, III, and IV only

Q6 Simplify F= (A'B'C)+ (AB'C) + (A'BC)+ (ABC) + (A'BC') using a K-map

(A) C'+AB (B) C+A'B'

(C) C+AB (D) C+A'B

Q7 The Karnaugh map for a Boolean function is given as

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	0	0	1	0
AB	1	1	1	1
$A\bar{B}$	0	1	1	1

The simplified Boolean equation for the above Karnaugh Map is

(A) AB+ CD+ AB'+AD

(B) AB + AC + AD + BCD

(C) AB + AD + BC + ACD

(D) AB + AC + BC + BCD

Q8 Minimize the following boolean function-

F(A, B, C, D) = Σm(3, 4, 5, 7, 9, 13, 14, 15)

(A) A'B+ AC'D + ABC

(B) A'BC' + A'CD + A'C'D'

(C) A'B + A'CD + AC' + ABC

(D) A'BC' + A'CD + AC'D + ABC

Q9 For the boolean expression $F = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}\,B\overline{C} + A\overline{B}\,\overline{C}$, the $+ ABC + AB\overline{C}$ minimized Product of Sum (POS) expression is

(A) $F = (\overline{B} + C)(\overline{A} + C)$

(B) $F = (\overline{B} + C)(A + \overline{C})$

(C) $F = (B + \overline{C})(A + \overline{C})$

(D) $F = \overline{C}(A + B + C)$

Q10 Simplify F= A'B'C+A'BC using a K-map

(A) B'C (B) A'C

(C) A'BC (D) AC

Answer Key

Q1	C
Q2	B
Q3	C
Q4	A
Q5	A
Q6	D
Q7	B
Q8	D
Q9	C
Q10	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

The number of cells required in a Karnaugh Map (K-map) for a Boolean expression consisting of n variables is determined by the number of possible combinations of those variables.

Number of Variables: If a Boolean expression has n variables, each variable can be either 0 or 1.

Number of Possible Combinations: The number of different combinations of n variables is 2^n

Number of Cells in K-Map: Each combination corresponds to a cell in the K-map.

Thus, the number of cells required in a Karnaugh Map for a Boolean expression with n variables is 2^n

Examples

For 2 Variables: n=2

Number of cells = 2^2=4

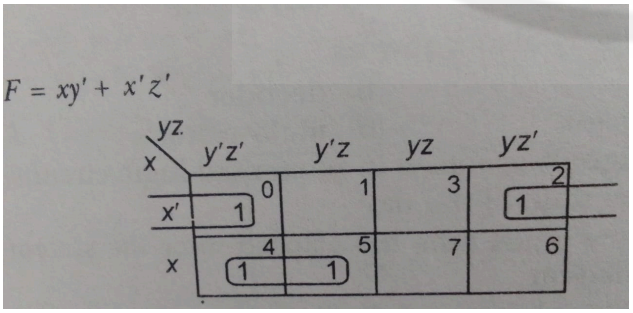
K-map will have 4 cells.

For 3 Variables: n=3

Number of cells = 2^3=8

K-map will have 8 cells.

Q2 Text Solution:



There are two pairs xy'z' + xy'z = xy'(z'+z) = xy' and x'y'z'+ x'yz' = x'z'(y'+y) = x'z'

So the simplified form will be xy' + x'z'

Q3 Text Solution:

Option A) This involves applying Boolean algebra laws and theorems to simplify expressions manually.

Key Laws:

Identity Law:

A 1=A and

A+0=A

Null Law:

A 0=0 and

A+1=1

Complement Law:

A A'=0 and

A+A' =1

Distributive Law:

A (B+C)=(A B)+(A C)

De Morgan's Theorems:

(A B)' = A' + B'

and

(A+B)' =A'.B'

Option B) Karnaugh Maps (K-Maps): A graphical method for simplifying Boolean expressions by grouping ones (or zeros) in a grid format. It is especially useful for simplifying expressions.

Q4 Text Solution:

K-map can take two forms:

1. Sum of product (SOP): The Sum of Products form is a Boolean expression in which several product terms (AND operations) are summed (OR operations) together. e.g. A'BC+A'B'C
2. Product of Sum (POS): The Product of Sums form is a Boolean expression where several sum terms (OR operations) are multiplied (AND operations) together. e.g. (A+B) (A'+C)

Q5 Text Solution:

Statement I: "We can either group 0's with 0's or 1's with 1's but we can not group 0's and 1's together."

True. In Karnaugh Maps, we can group cells containing 1's to simplify expressions (for SOP forms) or cells containing 0's to simplify expressions (for POS forms). You cannot mix 0's and 1's in the same group. Each group must be homogenous (all 1's or all 0's).

Statement II: "X representing don't care can be grouped with 0's as well as 1's."

True. In K-Maps, cells with a don't care condition (represented by X) can be grouped with both 0's and 1's. Don't care conditions provide flexibility in grouping, as they can be used to simplify the expression further by being included in either the 0-groups or 1-groups based on what is most beneficial for simplification.

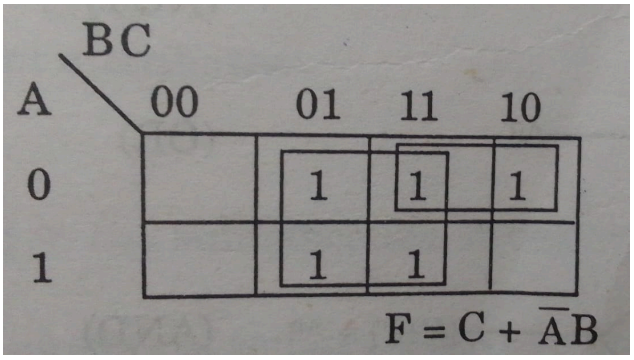
Statement III: "Groups may overlap each other and a group can only contain 2^n i.e. 1, 2, 4, 8, 16, and so on a number of cells."

True. Groups in a K-Map can overlap. The size of each group must be a power of 2 (1, 2, 4, 8, 16, etc.) to ensure that the grouping adheres to the principles of Boolean algebra simplification. This allows the simplification of the Boolean expression into a minimal number of terms.

Statement IV: "We can create groups of diagonal or any other shape."

False. In K-Maps, groups must be rectangular (or square) and adhere to the K-map grid. Diagonal groups or other non-rectangular shapes are not allowed. The grouping must be done in powers of 2 and must follow the grid structure to maintain the adjacency of cells.

Q6 Text Solution:



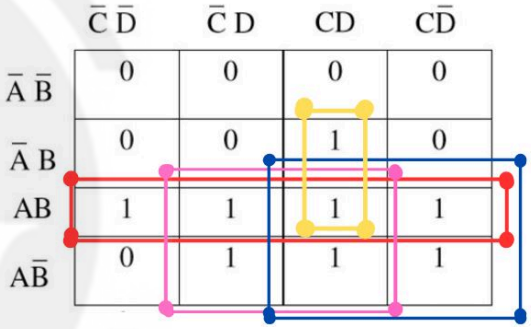
Here one quad is there (A'B'C)+ (AB'C) + (A'BC)+ (ABC) = B'C(A+A') + BC(A'+A) = B'C+BC = C

One pair is there = A'BC + A'BC' = A'B (C+C') = A'B

So simplified form will be = C+A'B

Q7 Text Solution:

Explanation:

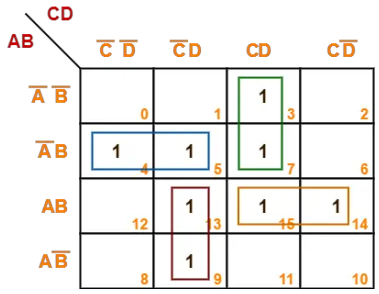


By grouping we will simply get AB + AC + AD + BCD.

The Final Answer is: Option B

Q8 Text Solution:

Since the given boolean expression has 4 variables, so we draw a 4 x 4 K Map. We fill the cells of the K Map in accordance with the given boolean function. Then, we form the groups in accordance with the above rules



F(A, B, C, D)

= A'B(C'D' + C'D) + (A'B' + A'B)(CD) + (AB + AB')(C'D) + AB(CD + CD')

= A'BC' + A'CD + AC'D + ABC

Thus, the minimized boolean expression is-

F(A, B, C, D) = A'BC' + A'CD + AC'D + ABC

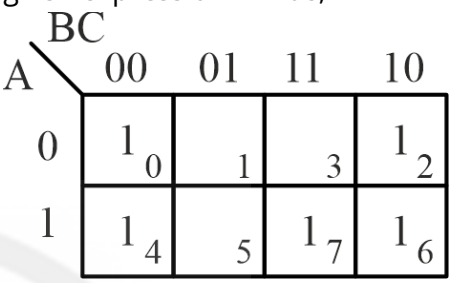
Q9 Text Solution:

Calculation:

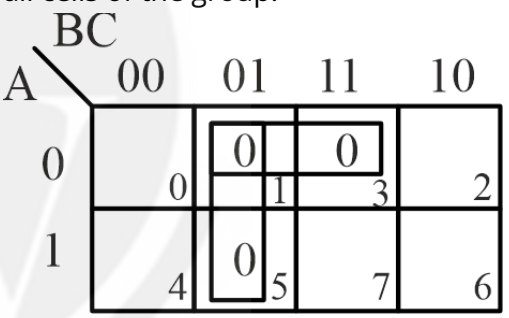
The given boolean expression is,

$$F = \overline{A}\ \overline{B}\ \overline{C} + \overline{A}\,B\,\overline{C} + A\,\overline{B}\ \overline{C} + ABC + AB\,\overline{C}$$

This is in SOP form. Now, since there are 3 variables A,B,C So, the number of cells in the K-map will be $2^n = 2^3 = 8$. The K-map for the given expression will be,



The minimized POS can be drawn using the K-map by just grouping the 0's and a variable appears in true form if it is 1 in all cells of the group, and in complemented form if it is 0 in all cells of the group.

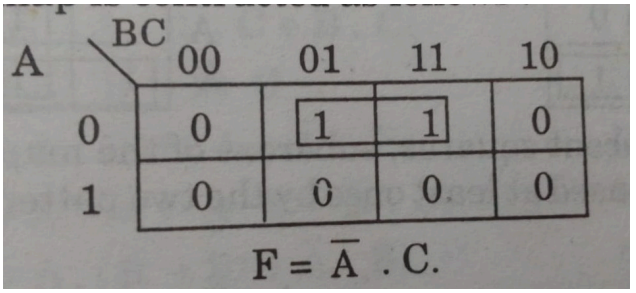


The minimized boolean function in the form of POS form will be,

$$F = (A + \overline{C})(B + \overline{C})$$

Thus, the correct option is (c).

Q10 Text Solution:



Here only one pair is there A'B'C + A'BC = A'C (B+B') = A'.C